

CARROLL COUNTY REGIONAL, MD, USA

WMO: 720699

Lat: 39.608N

Lon: 77.008W

Elev: 241

StdP: 98.47

Time Zone: -5.00 (NAE)

Period: 10-19

WBAN: 03724

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.6	-10.3	-22.0	0.5	-8.8	-18.8	0.7	-6.8	10.7	0.5	9.3	-2.0	3.4	290	0.425

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	34.0	24.1	32.7	23.6	32.0	23.5	26.8	31.1	25.4	30.6	24.6	29.6	3.1	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.9	21.9	29.4	23.9	19.3	28.3	22.8	18.1	27.1	85.5	31.5	79.1	31.4	75.6	28.9	32.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
8.8	7.7	6.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
			DB												
			WB	-15.7	28.6	3.3	1.9	-18.1	29.9	-20.0	31.0	-21.9	32.1	-24.3	33.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.4	-0.4	2.1	6.1	13.1	19.1	23.2	26.0	24.3	20.8	14.4	7.5	3.3	(6)	
	DBStd	10.16	5.67	6.47	5.73	5.28	4.75	3.19	2.85	2.77	3.89	4.67	4.61	5.10	(7)	
	HDD10.0	1059	325	231	149	26	1	0	0	0	0	12	100	216	(8)	
	HDD18.3	2593	581	454	379	173	51	3	0	0	21	140	325	466	(9)	
	CDD10.0	2290	2	11	30	119	283	397	497	444	324	148	26	9	(10)	
	CDD18.3	783	0	1	1	16	76	151	239	187	94	18	1	0	(11)	
	CDH23.3	6656	0	4	8	142	627	1287	2311	1512	658	107	1	0	(12)	
	CDH26.7	2547	0	0	0	38	217	467	1020	562	218	24	0	0	(13)	
(14) Wind	WSAvg	2.6	3.2	3.1	3.2	3.2	2.4	2.3	1.8	1.6	1.8	2.6	2.9	2.8	(14)	
(15) Precipitation	PrecAvg	1054	77	63	95	89	107	97	99	89	95	80	84	85	(15)	
	PrecMax	1583	204	136	194	196	228	463	238	224	446	210	185	196	(16)	
	PrecMin	755	16	11	34	24	13	10	9	25	20	2	10	8	(17)	
	PrecStd	207	39	34	40	43	49	77	53	50	84	46	43	51	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.0	22.4	24.1	30.9	32.9	34.2	37.2	35.0	33.0	29.1	22.3	18.9	(19)	
		MCWB	13.4	16.0	14.6	17.6	23.5	23.5	25.5	24.1	22.5	21.8	16.7	17.3	(20)	
	2%	DB	12.8	17.5	21.2	27.4	31.3	32.7	34.9	32.8	31.4	26.3	18.9	17.0	(21)	
		MCWB	10.0	12.4	14.4	17.3	22.9	23.7	25.5	23.1	22.4	19.5	14.7	15.5	(22)	
	5%	DB	10.2	14.0	17.9	24.0	28.9	31.2	32.9	32.0	29.1	23.7	17.1	13.0	(23)	
		MCWB	7.8	10.1	12.6	15.7	21.4	23.1	24.3	22.7	21.8	18.8	12.8	11.1	(24)	
	10%	DB	7.4	11.2	15.8	22.1	27.3	29.1	32.1	29.9	27.3	22.1	14.2	10.9	(25)	
		MCWB	5.1	7.7	10.8	15.2	20.1	22.0	24.3	21.9	20.4	17.6	10.2	9.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.8	17.2	17.8	21.3	27.3	27.3	29.6	27.7	25.2	23.2	17.9	17.6	(27)	
		MCDB	15.4	21.9	21.2	26.5	29.9	30.5	32.9	31.3	30.7	26.3	19.6	18.7	(28)	
	2%	WB	10.6	14.0	15.7	19.2	23.8	25.4	27.6	25.2	24.1	21.7	15.8	15.3	(29)	
		MCDB	12.2	15.9	19.6	23.8	29.1	29.8	32.3	30.6	28.8	25.0	18.0	16.8	(30)	
	5%	WB	7.8	10.5	13.8	17.7	22.6	24.2	25.9	24.2	23.0	20.0	13.9	12.1	(31)	
		MCDB	9.4	13.8	17.1	22.5	27.7	29.1	31.0	29.2	27.5	23.1	16.7	13.1	(32)	
	10%	WB	5.4	7.7	11.2	16.0	21.4	23.3	25.1	23.3	21.9	18.1	11.8	8.6	(33)	
		MCDB	7.5	9.6	14.9	20.7	26.0	28.0	30.5	27.4	25.5	20.6	13.6	9.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.7	8.7	9.5	11.4	10.6	10.8	10.6	10.4	10.2	9.4	8.6	6.9	(35)	
		MCDBR	11.1	12.9	13.1	14.1	12.9	12.3	12.0	12.0	12.4	11.6	11.1	9.5	(36)	
	5% WB	MCWBR	10.0	9.2	8.3	7.2	6.4	5.6	4.9	5.3	5.6	6.2	7.3	8.5	(37)	
		MCWBR	10.2	11.7	12.4	13.1	12.2	10.9	10.7	10.4	10.1	9.1	9.2	9.0	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.304	0.315	0.353	0.394	0.430	0.492	0.507	0.501	0.420	0.357	0.318	0.302	(40)	
		taud	2.447	2.395	2.369	2.276	2.206	2.070	2.044	2.045	2.245	2.427	2.498	2.488	(41)	
	Ebn at Noon		867	905	902	881	852	797	780	773	819	845	844	842	(42)	
		Edh at Noon	80	97	111	130	143	164	166	162	125	93	76	72	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.93	2.81	3.73	4.80	5.36	5.95	5.92	5.23	4.27	3.12	2.23	1.68	(44)	
	RadStd		0.17	0.33	0.25	0.40	0.62	0.46	0.32	0.28	0.54	0.35	0.19	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (73 neighbors)	+0.42	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+121	+67

Nomenclature: See separate page